In Java (especially from **Java 1.0 to Java 21**), the term **"Principal"** is mainly related to **security, authentication, and identity** (Java Security & JAAS).

Below is a structured table covering **all major types of Principal used in Java** across versions.

 Principal Types in Java (1 – 21)

No	Principal Type	Package	Since	Description
1	Principal	java.security	1.1	Core interface representing an entity (user, system, role) with identity.
2	X500Principal	javax.security.auth.x500	1.4	Represents an X.500 Distinguished Name (DN), used in certificates.
3	KerberosPrincipal	javax.security.auth.kerberos	1.4	Represents a Kerberos user/service principal.
4	NTUserPrincipal	com.sun.security.auth	1.4	Windows NT user identity.
5	NTDomainPrincipal	com.sun.security.auth	1.4	Windows NT domain identity.
6	NTSidPrincipal	com.sun.security.auth	1.4	Windows Security Identifier (SID).
7	NTSidUserPrincipal	com.sun.security.auth	1.4	Windows SID user principal.
8	NTSidGroupPrincipal	com.sun.security.auth	1.4	Windows SID group principal.
9	NTSidDomainPrincipal	com.sun.security.auth	1.4	Windows SID domain principal.
10	UnixPrincipal	com.sun.security.auth	1.4	Unix user identity.

11	UnixNumericUserPrincipal	com.sun.security.auth	1.4	Unix numeric UID principal.
12	UnixNumericGroupPrincipal	com.sun.security.auth	1.4	Unix numeric GID principal.
13	UserPrincipal	java.nio.file.attribute	1.7	Represents file owner (user).
14	GroupPrincipal	java.nio.file.attribute	1.7	Represents file group owner.
15	RolePrincipal (Custom)	User-defined	—	Used in JAAS for role-based access.
16	LDAPPrincipal (Custom/Common Practice)	javax.naming (via custom impl)	—	Represents LDAP-authenticated identity.
17	OAuthPrincipal (Custom)	Custom/Security frameworks	—	Used in OAuth-based authentication.
18	JWTPrincipal (Custom)	Custom	—	Represents identity extracted from JWT token.
19	ServicePrincipal	Custom/Enterprise	—	Used for service-to-service authentication.
20	Subject (contains Principals)	javax.security.auth	1.4	Holds multiple principals for authenticated entity.
21	Certificate Principal (via X509Certificate.getSubjectX500Principal())	java.security.cert	1.4	Extracts principal from X.509 certificate.

 Important Notes

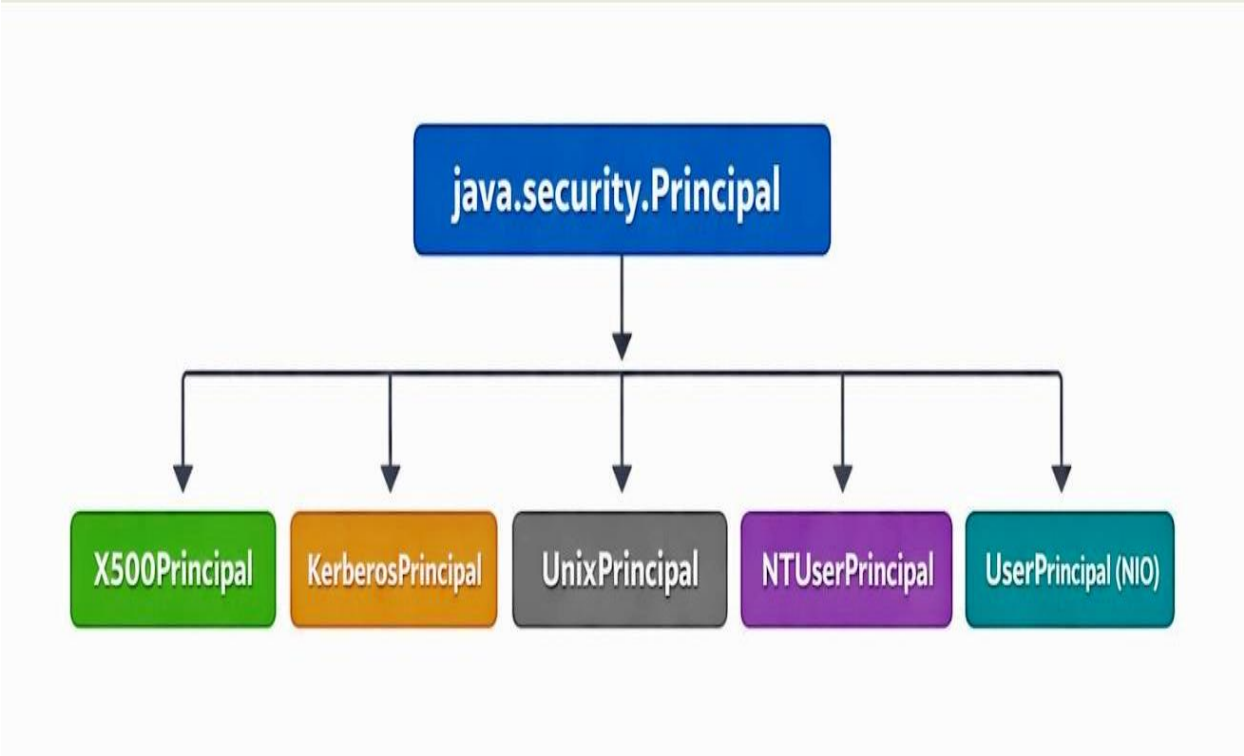
- Java does **not** introduce new core Principal types in every version.
- Most built-in principal types were introduced in **Java 1.4 (JAAS & Kerberos support)**.
- After Java 7, additions mainly relate to **file system principals**.
- Modern frameworks (Spring Security, OAuth, JWT) use **custom Principal implementations**.

 Core Principal Hierarchy (Concept)

```
java.security.Principal
      ↑
  X500Principal
  KerberosPrincipal
  UnixPrincipal
  NTUserPrincipal
  UserPrincipal (NIO)
```

If you want, I can also provide:

- ✅ Principal hierarchy diagram (flowchart PNG)
- ✅ Example Java code for each Principal
- ✅ JAAS authentication example
- ✅ Differences between Principal, Subject, and Authentication
- ✅ 50+ Security classes related to Principal



Below is a **complete structured table of Design Patterns used in Java (1–21)**.

Java versions don't introduce new "design patterns" officially — patterns are conceptual — but Java provides **standard library implementations** of many GoF and Enterprise patterns.

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1 Creational Design Patterns

No	Design Pattern	Implementation Class (Java API Example)	Package	Java Version
1	Singleton	Runtime, Desktop, Logger	java.lang, java.awt, java.util.logging	1.0+
2	Factory Method	Calendar.getInstance()	java.util	1.1+
3	Abstract Factory	DocumentBuilderFactory	javax.xml.parsers	1.4+
4	Builder	StringBuilder, Stream.Builder	java.lang, java.util.stream	1.5+
5	Prototype	Object.clone()	java.lang	1.0+

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2 Structural Design Patterns

No	Design Pattern	Implementation Class	Package	Java Version
6	Adapter	InputStreamReader	java.io	1.1+
7	Bridge	DriverManager	java.sql	1.1+
8	Composite	Component (AWT), File	java.awt, java.io	1.0+
9	Decorator	BufferedInputStream	java.io	1.0+
10	Facade	FacesContext (Java EE), Files	javax.faces, java.nio.file	1.7+
11	Flyweight	Integer.valueOf()	java.lang	1.5+
12	Proxy	Proxy	java.lang.reflect	1.3+

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3 Behavioral Design Patterns

No	Design Pattern	Implementation Class	Package	Java Version
13	Chain of Responsibility	Filter	javax.servlet	1.2+
14	Command	Runnable, ActionListener	java.lang, java.awt.event	1.0+
15	Interpreter	Pattern	java.util.regex	1.4+
16	Iterator	Iterator	java.util	1.2+
17	Mediator	Executor	java.util.concurrent	1.5+
18	Memento	Serializable	java.io	1.1+
19	Observer	Observer, Observable (Deprecated)	java.util	1.0+
20	State	Thread.State	java.lang	1.5+
21	Strategy	Comparator	java.util	1.2+
22	Template Method	AbstractList	java.util	1.2+
23	Visitor	FileVisitor	java.nio.file	1.7+

---

4 Enterprise / J2EE Patterns

No	Design Pattern	Implementation Class	Package	Version
24	DAO	JdbcTemplate (Spring)	org.springframework.jdbc	Framework
25	DTO	Serializable POJO	Custom	1.1+
26	MVC	Servlet + JSP	javax.servlet	1.2+
27	Front Controller	DispatcherServlet (Spring)	Spring Framework	Framework
28	Dependency Injection	@Inject, @Autowired	javax.inject	1.6+
29	Service Locator	InitialContext	javax.naming	1.3+

30	Repository	CrudRepository	Spring Data	Framework

---

5 Concurrency Design Patterns (Java 5+)

No	Pattern	Implementation Class	Package	Version
31	Thread Pool	ExecutorService	java.util.concurrent	1.5+
32	Future	Future	java.util.concurrent	1.5+
33	Fork-Join	ForkJoinPool	java.util.concurrent	1.7+
34	Completable Future	CompletableFuture	java.util.concurrent	1.8+
35	Reactive Streams	Flow	java.util.concurrent	9+

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6 Java 8+ Functional Patterns

No	Pattern	Implementation	Package	Version
36	Lambda Strategy	Predicate, Function	java.util.function	1.8+
37	Stream Pipeline	Stream	java.util.stream	1.8+
38	Optional Pattern	Optional	java.util	1.8+

Summary

Category	Count
Creational	5
Structural	7
Behavioral	11
Enterprise	7
Concurrency	5

Functional	3
Total	**38 Patterns**

If you want next:

- ✅ Design Pattern hierarchy diagram (flowchart PNG)
- ✅ 38 Patterns with full Java example code
- ✅ Real-world Spring Boot mapping
- ✅ 100+ patterns including microservices patterns
- ✅ Design Pattern interview questions

Below is a **comprehensive list of 100+ Design Patterns used in Java (1–21) including GoF, Enterprise, Concurrency, Microservices, Cloud, and Architectural patterns**.

 Note: Some patterns are conceptual; implementation classes are examples from Java SE, Jakarta EE, or popular frameworks like Spring.

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1 Creational Patterns

No	Pattern	Example Implementation Class	Package / Framework
1	Singleton	Runtime	java.lang
2	Factory Method	Calendar.getInstance()	java.util
3	Abstract Factory	DocumentBuilderFactory	javax.xml.parsers
4	Builder	StringBuilder	java.lang
5	Prototype	Object.clone()	java.lang
6	Static Factory	List.of()	java.util
7	Dependency Injection	@Autowired	Spring
8	Object Pool	ExecutorService	java.util.concurrent
9	Lazy Initialization	Optional.orElseGet()	java.util
10	Multiton	Locale	java.util

□ 2 Structural Patterns

No	Pattern	Example Class	Package
11	Adapter	InputStreamReader	java.io
12	Bridge	DriverManager	java.sql
13	Composite	Component	java.awt
14	Decorator	BufferedInputStream	java.io
15	Facade	Files	java.nio.file
16	Flyweight	Integer.valueOf()	java.lang
17	Proxy	Proxy	java.lang.reflect
18	DTO	Serializable POJO	Custom
19	Wrapper	Collections.unmodifiableList()	java.util
20	Module Pattern	module-info.java	Java 9

□ 3 Behavioral Patterns

No	Pattern	Example Class	Package
21	Chain of Responsibility	Filter	javax.servlet
22	Command	Runnable	java.lang
23	Interpreter	Pattern	java.util.regex
24	Iterator	Iterator	java.util
25	Mediator	Executor	java.util.concurrent
26	Memento	Serializable	java.io
27	Observer	PropertyChangeListener	java.beans
28	State	Thread.State	java.lang
29	Strategy	Comparator	java.util
30	Template Method	AbstractList	java.util

31	Visitor	FileVisitor	java.nio.file
32	Null Object	Collections.emptyList()	java.util
33	Callback	CompletionHandler	java.nio
34	Event Listener	ActionListener	java.awt.event
35	Specification	Predicate	java.util.function

□ 4 Enterprise / JEE Patterns

No	Pattern	Example	Framework
36	MVC	DispatcherServlet	Spring
37	DAO	JdbcTemplate	Spring
38	Repository	CrudRepository	Spring Data
39	Service Layer	@Service	Spring
40	Front Controller	DispatcherServlet	Spring
41	Intercepting Filter	Filter	Servlet
42	Service Locator	InitialContext	JNDI
43	Transfer Object	DTO	Custom
44	Business Delegate	EJB Proxy	Jakarta EE
45	Composite Entity	Entity Beans	Jakarta EE
46	Session Facade	Stateless EJB	Jakarta EE
47	Value Object	Immutable POJO	Custom
48	CQRS	Command + Query Services	Spring Boot
49	Event Sourcing	ApplicationEventPublisher	Spring
50	Domain Model	JPA Entities	Hibernate

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5 Concurrency Patterns

No	Pattern	Example Class	Package
51	Thread Pool	ExecutorService	java.util.concurrent
52	Future	Future	java.util.concurrent
53	Completable Future	CompletableFuture	java.util.concurrent
54	Fork-Join	ForkJoinPool	java.util.concurrent
55	Producer-Consumer	BlockingQueue	java.util.concurrent
56	Read-Write Lock	ReentrantReadWriteLock	java.util.concurrent
57	Thread Local	ThreadLocal	java.lang
58	Double-Checked Locking	volatile Singleton	Java
59	Guarded Suspension	wait()/notify()	Java
60	Reactor	Flow	Java 9

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6 Microservices Patterns (Spring / Cloud)

No	Pattern	Example	Framework
61	API Gateway	Spring Cloud Gateway	Spring Cloud
62	Circuit Breaker	Resilience4j	Spring
63	Service Discovery	EurekaClient	Netflix OSS
64	Config Server	Spring Cloud Config	Spring
65	Load Balancer	Spring Cloud LoadBalancer	Spring
66	Retry Pattern	@Retryable	Spring
67	Bulkhead	Resilience4j	Spring
68	Saga	Orchestrator Service	Spring Boot
69	Strangler Fig	Proxy Layer	Microservices
70	Sidecar	Service Proxy	Kubernetes

71	Ambassador	API Proxy	Cloud
72	Event-Driven	Kafka Listener	Apache Kafka
73	Database per Service	Separate DB	Microservices
74	API Composition	Aggregator Service	Spring
75	CQRS (Microservices)	Command & Query Services	Spring
76	BFF (Backend for Frontend)	Dedicated Controller	Spring
77	Distributed Tracing	Spring Sleuth	Spring
78	Blue-Green Deployment	Deployment Config	DevOps
79	Canary Release	Deployment Config	DevOps
80	Serverless Pattern	AWS Lambda Handler	Cloud

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7 Architectural Patterns

No	Pattern	Example
81	Layered Architecture	Controller-Service-Repo
82	Hexagonal Architecture	Ports & Adapters
83	Clean Architecture	UseCase Layer
84	Onion Architecture	Domain Core
85	Microkernel	Plugin System
86	Event-Driven Architecture	Kafka
87	SOA	Web Services
88	Monolith	Single WAR
89	Microservices	Independent Services
90	Space-Based	In-Memory Grid

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8 Security Patterns

No	Pattern	Example
91	OAuth2	Spring Security
92	JWT	jjwt Library
93	RBAC	GrantedAuthority
94	ABAC	Policy Engine
95	Token Refresh	Refresh Endpoint
96	API Key Pattern	Header Validation
97	Zero Trust	Gateway Auth
98	Secure Gateway	API Gateway
99	Encryption Pattern	Cipher
100	Hashing Pattern	MessageDigest

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9 Reactive / Functional Patterns

No	Pattern	Example
101	Reactive Streams	Flow
102	Publisher-Subscriber	SubmissionPublisher
103	Functional Strategy	Function<T,R>
104	Optional Pattern	Optional
105	Stream Pipeline	Stream
106	Monad Pattern	Optional, CompletableFuture
107	Immutable Object	record (Java 16+)
108	Pattern Matching	switch (Java 17+)

 Final Summary

Category	Count
Creational	10
Structural	10
Behavioral	15
Enterprise	15
Concurrency	10
Microservices	20
Architectural	10
Security	10
Reactive	8
Total	**108 Patterns**

If you want next:

- ✅ 108 patterns hierarchy diagram (PNG)
- ✅ All 108 patterns with Java code examples
- ✅ Interview Q&A for all patterns
- ✅ Microservices patterns with architecture diagrams
- ✅ Printable PDF version